

SOC SOAR Capstone Project – Malware Analysis to Incident Response

Case Management, SOAR, Threat Intelligence & Incident Response

Project Continuation Overview (Linked to Part 1)

This document represents **Part 2** of the SOC Capstone Project and is a **direct continuation of Part 1**, where a **realistic Windows malware attack** was simulated on an AWS Windows EC2 instance and detected using **Wazuh + Sysmon**.

While **Part 1** focused on:

- Malware execution simulation
- Endpoint telemetry collection
- Alert generation and correlation in Wazuh

Part 2 focuses on the SOC analyst response phase, covering **case management, investigation, response, automation, and reporting** using a **SOAR-driven workflow**.

Part 2 Objective

The objective of this phase is to demonstrate how a SOC analyst:

- Converts alerts into an **incident case**
- Performs **structured investigation**
- Uses **SOAR automation** for enrichment
- Executes **full incident response lifecycle**
- Shares validated intelligence via **MISP**
- Produces **final incident reporting and metrics**

This reflects **real-world SOC operations**, not lab-only detection.

Tools & Platforms Used (SOAR Stack)

Wazuh (SIEM / XDR)

- Source of alerts, telemetry, and correlation
- Sysmon-based behavioral detection
- MITRE ATT&CK mapped alerts

TheHive (Case Management & IR)

- Alert triage and case creation
- Task-based incident response workflow

- Observables, TTPs, timeline, reporting

Cortex (SOAR Automation)

- Automated observable enrichment
- VirusTotal, OTX, MISP analyzers
- Evidence-backed decision making

MISP (Threat Intelligence Platform)

- IOC sharing and correlation
- MITRE Galaxy mapping
- Intelligence lifecycle integration

SOAR & Automation Context

This project demonstrates **SOAR in action**, including:

- Alert → Case automation
- Observable enrichment via Cortex
- MITRE ATT&CK technique mapping
- IOC validation before intelligence sharing
- Reduced manual effort with analyst control

Automation is **assisted**, not blind — analyst judgment is applied at every stage.

SOC Analyst Role Demonstrated

In Part 2, the analyst performs:

- Alert triage and validation
- Case creation and scoping
- Observable extraction and enrichment
- Threat classification (True Positive)
- Incident response execution
- Threat intelligence sharing
- Incident reporting and metrics review

This reflects a **Tier 1 / Tier 2 SOC analyst workflow**.

Incident Scope Summary (From Part 1)

The incident under investigation involves:

- PowerShell-based malware execution
- External payload retrieval

- Malicious file drop
- Registry-based persistence
- Confirmed external communication
- No lateral movement or data exfiltration

Part 2 begins **after these activities were confirmed in Wazuh.**

Architecture Overview (Logical Flow)

Endpoint (Windows EC2)

↓

Sysmon Telemetry

↓

Wazuh Detection & Correlation

↓

TheHive Alert Intake

↓

Case Creation & Investigation

↓

Cortex Automated Enrichment

↓

MISP Threat Intelligence Sharing

↓

Incident Closure & Reporting

Why This Part Matters

Most portfolios stop at **alert detection**.

This part proves:

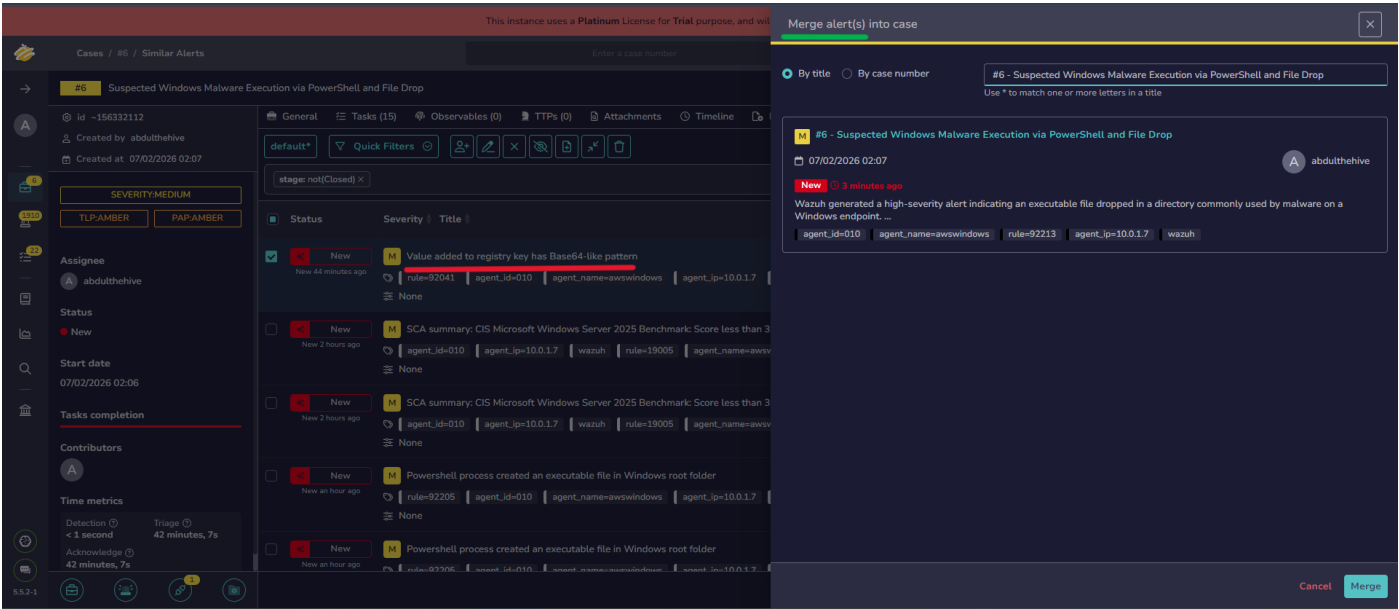
- I can **run an incident**
- I understand **IR frameworks**
- I can **document and close cases**
- I can **share intelligence responsibly**
- I can work with **SOAR tooling**

Step-by-Step Execution (Continuation of Part 1)

Step 12: Merge Related Alerts into the Case

The registry modification alert (“Value added to registry key has Base64-like pattern”) was generated immediately after the malicious PowerShell execution.

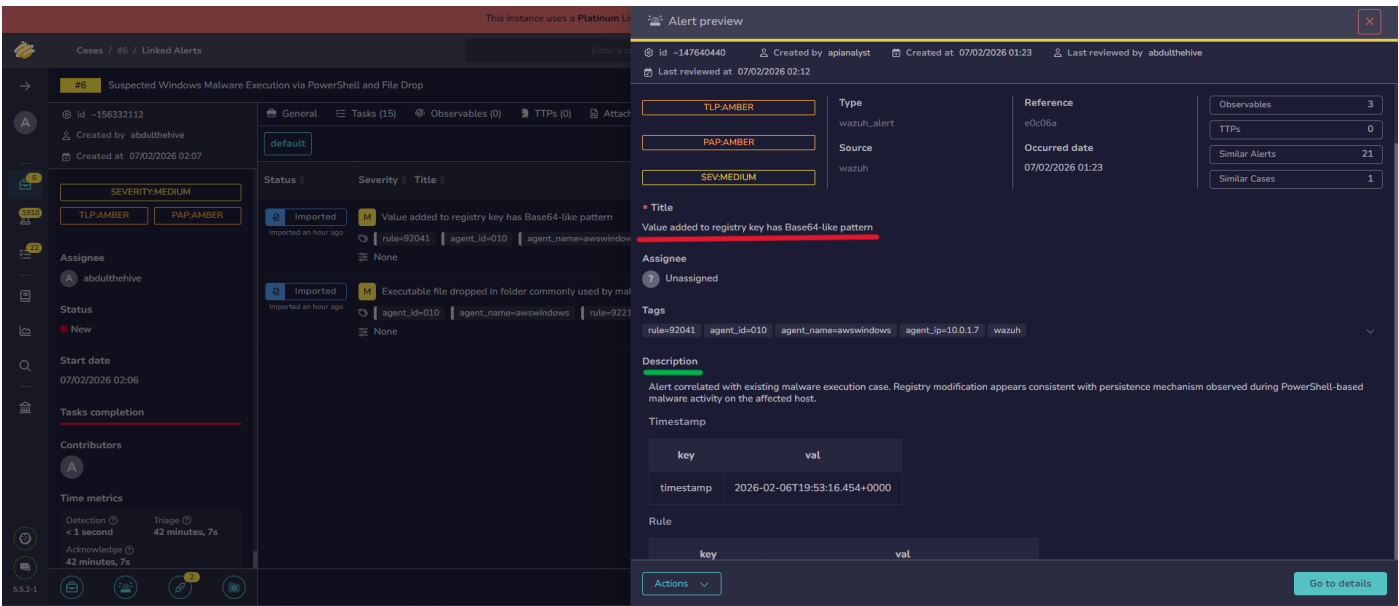
Since both alerts occurred on the same host, within the same timeframe, and are part of the same attack chain, the alert was merged into the existing case.



Step 13: Review Merged Alert Description

After merging, the alert details were reviewed to confirm relevance.

The rule indicated registry modification consistent with persistence behavior, validating that this alert belongs to the same malware execution sequence.



Step 14: Review Case Alert Details (Primary Alert)

The main case alert was examined to identify key fields for investigation:

- Executing user
- PowerShell process details
- Target file path

Command line used

These fields were marked for conversion into observables.

~156332112

ated by **abdulthehive**

ated at 07/02/2026 02:07

dated at 07/02/2026 02:09

SEVERITY:MEDIUM

PAP:AMBER

PAP:AMBER

nee

odulthehive

s

Y

date

2026 02:06

completion

butors

General

Tasks (15)

Observables (2)

TTPs (0)

Attachments

Timeline

Report

Rule

key	val
rule.level	15
rule.description	Executable file dropped in folder commonly used by malware
rule.id	92213
rule.mitre.id	<u>[T1105]</u>
rule.mitre.tactic	<u>[Command and Control]</u>
rule.mitre.technique	<u>[Ingress Tool Transfer]</u>
rule.firedtimes	6
rule.mail	True
rule.groups	[sysmon', 'sysmon_eid11_detections', 'windows']

Agent

Suspected Windows Malware Execution via PowerShell and File Drop

156332112

Created by: abdulthehive

Created at: 07/02/2026 02:07

Updated at: 07/02/2026 02:09

SEVERITY:MEDIUM

AMBER

PAP:AMBER

Created by: abdulthehive

Created at: 07/02/2026 02:07

Completion

Alerts

Metrics

On Triage 42 minutes, 7s

GeneralTasks (15)Observables (2)TTPs (0)AttachmentsTimelineReportPagesHistorySimilar alerts

CreationUtcTime: 2026-02-06 19:53:11.508	
User: EC2AMAZ-3NK44OT\Administrator	
data.win.eventdata.ruleName	technique_id=T1059.001,technique_name=PowerShell
data.win.eventdata.utcTime	2026-02-06 19:53:11.508
data.win.eventdata.processGuid	{03046e1c-46a7-6986-d803-000000002e00}
data.win.eventdata.processId	704
data.win.eventdata.image	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
data.win.eventdata.targetFilename	C:\Users\Administrator\AppData\Local\Temp\Z__PSScriptPolicyTest_4gix15h5.vwpg.ps1
data.win.eventdata.creationUtcTime	2026-02-06 19:53:11.508
data.win.eventdata.user	EC2AMAZ-3NK44OT\Administrator

Location

key	val
location	EventChannel

Merged with alert #e0c06a. Value added to registry key has Base64-like pattern

Step 15: Review Merged Alert Details (Registry Event)

The merged alert was analyzed in detail:

- Registry Run key modification detected
 - MITRE techniques mapped (Defense Evasion / Modify Registry)
 - Confirms persistence attempt following execution
- This strengthened confidence in malicious intent.

2112

abdulthehive

07/02/2026 02:07

07/02/2026 02:09

VERITY:MEDIUM

R

PAP:AMBER

hive

06

tion

Triage ⓘ
42 minutes, 7s

22

2

GeneralTasks (15)Observables (2)TTPs (0)AttachmentsTimeline

Merged with alert #e0c06a Value added to registry key has Base64-like pattern

Timestamp

key	val
timestamp	2026-02-06T19:53:16.454+0000

Rule

key	val
rule.level	10
rule.description	Value added to registry key has Base64-like pattern
rule.id	92041
rule.mitre.id	['T1027', 'T1112']
rule.mitre.tactic	['Defense Evasion']
rule.mitre.technique	['Obfuscated Files or Information', 'Modify Registry']
rule.firedtimes	1
rule.mail	False
rule.groups	['sysmon', 'sysmon_eid1_detections', 'windows']

Suspected Windows Malware Execution via PowerShell and File Drop

32112

by abdulthehive

at 07/02/2026 02:07

at 07/02/2026 02:09

SEVERITY:MEDIUM

ER

PAP:AMBER

hehive

02:06

letion

22

2

42 minutes, 7s

GeneralTasks (15)Observables (2)TTPs (0)AttachmentsTimelineReportPagesHistorySimilar alerts

data.win.eventdata.ruleName	technique_id=T1012,technique_name=Query Registry
data.win.eventdata.utcTime	2026-02-06 19:53:15.408
data.win.eventdata.processGuid	{03046e1c-46ab-6986-d903-000000002e00}
data.win.eventdata.processId	2456
data.win.eventdata.image	C:\Windows\System32\reg.exe
data.win.eventdata.fileVersion	10.0.26100.5074 (WinBuild.160101.0800)
data.win.eventdata.description	Registry Console Tool
data.win.eventdata.product	Microsoft® Windows® Operating System
data.win.eventdata.company	Microsoft Corporation
data.win.eventdata.originalFileName	reg.exe
data.win.eventdata.commandLine	reg add HKCU\Software\Microsoft\Windows\CurrentVersion\Run /v svchost_update /t REG_SZ /d \"C:\\Users\\Public\\svchost-update\\svchost_update.scr\" /f
data.win.eventdata.currentDirectory	C:\\Sysmon\\
data.win.eventdata.user	EC2AMAZ-3NK44OT\\Administrator
data.win.eventdata.logonGuid	{03046e1c-3867-6986-2608-3e0000000000}

6 Suspected Windows Malware Execution via PowerShell and File Drop

-156332112

reated by abdulthehive

reated at 07/02/2026 02:07

updated at 07/02/2026 02:09

SEVERITY:MEDIUM

PLP:AMBER

PAP:AMBER

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t date

2/2026 02:06

ts completion

tributors

e metrics

ection

second

42 minutes, 7s

GeneralTasks (15)Observables (2)TTPs (0)AttachmentsTimelineReportPagesHistorySimilar alerts

data.win.eventdata.currentDirectory	C:\\Sysmon\\
data.win.eventdata.user	EC2AMAZ-3NK44OT\\Administrator
data.win.eventdata.logonGuid	{03046e1c-3867-6986-2608-3e0000000000}
data.win.eventdata.logonId	0x3e0826
data.win.eventdata.sessionId	2
data.win.eventdata.integrityLevel	High
data.win.eventdata.hashes	SHA1=3922B8843F27EEC9BED40F54C88D683340E61F8,MD5=F6E3559DDDDCCC843A12CFD50178C554,SHA256=EF37663B44AC66920C6F33694DEEA01ACB78AE3F3012884819373FA04C3EB5F0,IMPHASH=2DC6AE94704E0DBEAF5A2BB00BD87F0
data.win.eventdata.parentProcessGuid	{00000000-0000-0000-0000-000000000000}
data.win.eventdata.parentProcessId	1580

Location

key	val
location	EventChannel

Step 16: Add Observables to the Case

Based on alert analysis, the following observables were added individually:

- **Filename:** svchost_update.scr
 - **File (attachment):** Dropped executable
 - **Hash (SHA-256):** Extracted from Wazuh logs
 - **Domain:** testphp.vulnweb.com
 - **Command:** PowerShell Invoke-WebRequest execution
- All observables were tagged, marked as sighted, and assigned TLP/PAP appropriately.

 Adding an Observable ✕

* Type

filename

* Value

One observable per line ☒ 5 observable(s)

file-drop
windows
wazuh
sysmon

TLP

TLP: CLEAR TLP: GREEN **TLP: AMBER** TLP: AMBER+STRICT TLP: RED

PAP

PAP: CLEAR PAP: GREEN **PAP: AMBER** PAP: RED

Is IOC ☒

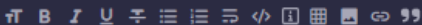
Has been sighted ☒

Ignore similarity ☒

Tags

malware file-drop

Description



Executable file dropped in C:\Users\Public\svchost-update as detected by Sysmon Event ID 11.

Preview ⓘ

Cancel

Save and add another

Confirm

Adding an Observable

Type

domain

Value

testphp.vulnweb.com

One observable per line1 observable(s)

TLP

TLP: CLEARTLP: GREENTLP: AMBERTLP: AMBER+STRICTTLP: RED

PAP

PAP: CLEARPAP: GREENPAP: AMPERPAP: RED

Is IOC

Has been sighted

Ignore similarity

Tags

networkc2-suspecteddns

Description

External domain contacted by PowerShell during suspected malware execution.

Cancel

Save and add another

Confirm

Adding an Observable

Type

other

Value

powershell -c Invoke-WebRequest http://testphp.vulnweb.com -UseBasicParsing

One observable per line1 observable(s)

TLP

TLP: CLEARTLP: GREENTLP: AMBERTLP: AMBER+STRICTTLP: RED

PAP

PAP: CLEARPAP: GREENPAP: AMPERPAP: RED

Is IOC

Has been sighted

Ignore similarity

Tags

powershellexecutiont1059

Description

PowerShell command observed during malicious activity.

Cancel

Save and add another

Confirm

Step 17: Run Cortex Analyzers (SOAR Automation)

Cortex analyzers were executed against the observables:

- VirusTotal
 - OTX
 - MISP lookup
 - Domain and hash reputation analyzers
- This demonstrates SOAR-based automated enrichment.

Analyzer

Other Analyzer

Select all / Deselect all

MISP_2_1 [Cortex]

Domain Analyzer

Select all / Deselect all

AIL_OnionLookup_1_0 [Cortex]

MISP_2_1 [Cortex]

OTXQuery_2_0 [Cortex]

VirusTotal_GetReport_3_1 [Cortex]

Hash Analyzer

Select all / Deselect all

CIRCLHashlookup_1_1 [Cortex]

MISP_2_1 [Cortex]

OTXQuery_2_0 [Cortex]

VirusTotal_GetReport_3_1 [Cortex]

VirusTotal_Rescan_3_1 [Cortex]

File Analyzer

Select all / Deselect all

ClamAV_FileInfo_1_1 [Cortex]

OTXQuery_2_0 [Cortex]

VirusTotal_GetReport_3_1 [Cortex]

VirusTotal_Scan_3_1 [Cortex]

Filename Analyzer

Select all / Deselect all

MISP_2_1 [Cortex]

Cancel

Run Selected Analyzers

Step 18: Review Analyzer Results

Analyzer results showed:

- No prior MISP sightings (expected in local simulation)
 - VirusTotal flagged the test domain as suspicious
 - Domain reputation confirmed external communication
- Results supported the malicious execution hypothesis.

The screenshot displays the 'Observables (7)' tab in a security tool. The interface shows a list of indicators of compromise (IOCs) with their associated data types, values, and analysis results. The table has columns for 'Flags', 'Data type', and 'Value/Filename'. The IOCs are listed as follows:

Flags	Data type	Value/Filename
TLP:AMBER PAP:AMBER	other	powershell -c Invoke-WebRequest hxxp://testphp[.]vulnweb[.]com -UseBasicParsing
	powershell	execution
		t1059
		MISP:Search="0 events"
TLP:AMBER PAP:AMBER	domain	testphp[.]vulnweb[.]com
	network	c2-suspected
		dns
		MISP:Search="0 events"
		VT:GetReport="1/93"
		VT:GetReport="3 resolution(s)"
		OTX:Pulses="Number of pulses: 0,..."
TLP:AMBER PAP:AMBER	hash	EF3763B4AAC66920CF33694DEEA01ACB78AE3F3012884819373FA04C3EB5F0
	hash	malware
		MISP:Search="0 events"
TLP:AMBER PAP:AMBER	file	svchost_update[.]scr
	persistence	suspicious-path
		VT:Scan="0/76"
		OTX:Pulses="Number of pulses: 0,..."
TLP:AMBER PAP:AMBER	filename	file-drop
	malware	file-drop
		MISP:Search="0 events"
TLP:AMBER PAP:AMBER	filename	svchost_update[.]scr malware
	malware	file-drop
		MISP:Search="0 events"

Step 19: Map MITRE ATT&CK TTPs

Relevant TTPs were added to the case:

- **T1059.001** – PowerShell
 - **T1105** – Ingress Tool Transfer
 - **T1112** – Modify Registry
 - **T1027** – Obfuscated Files or Information
- This formally mapped the attack chain to MITRE ATT&CK.

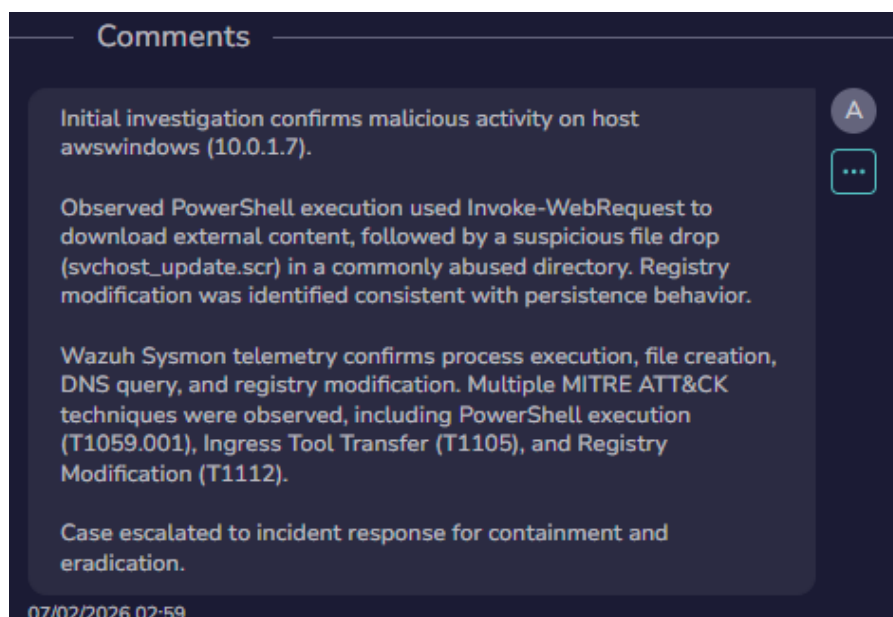
General Tasks (15) Observables (8) TTPs (4) Attachments Timeline Report Pages History Similar alerts Linked alerts Responders									
+ default Export list									
Tactic	Technique	By	Dates	O.	C.	U.			
execution	T1059.001 - PowerShell	A	O. 07/02/2026 01:23 C. 07/02/2026 02:51						
defense-evasion	T1112 - Modify Registry	A	O. 07/02/2026 01:24 C. 07/02/2026 02:53 U. 07/02/2026 02:55						
defense-evasion	T1027 - Obfuscated Files or Information	A	O. 07/02/2026 01:24 C. 07/02/2026 02:54 U. 07/02/2026 02:55						
command-and-control	T1105 - Ingress Tool Transfer	A	O. 07/02/2026 01:23 C. 07/02/2026 02:52						

Step 20: Analyst Comment – Case Confirmation

An analyst comment was added summarizing findings:

- PowerShell used to retrieve external content
- Suspicious file dropped in common malware directory
- Registry persistence observed
- Multiple Sysmon events correlated

Based on enrichment and correlation, the case was confirmed as a **malware execution incident**, and incident response tasks were initiated.



Step 21 – Start of Incident Response Tasks

Context / Transition Step

After confirming malicious activity through:

- PowerShell execution
- Suspicious file drop
- Registry-based persistence

- Correlated Sysmon telemetry

The case was **formally escalated to Incident Response**.

All response activities were executed using **TheHive task-driven workflow**, following standard IR phases:

Preparation → Detection & Analysis → Containment → Eradication → Recovery → Post-Incident Review

The following tasks document each response action performed, with evidence, screenshots, and analyst notes captured directly within the case timeline.

Task 1 – Preparation

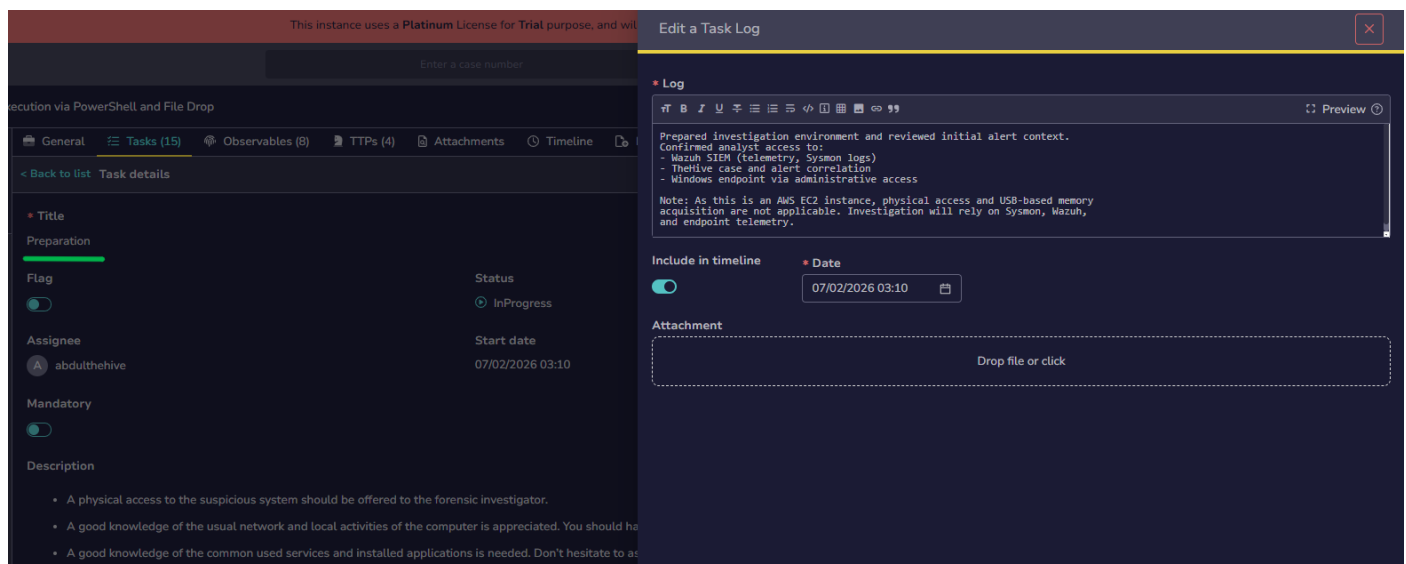
Phase: Preparation

What was done:

- Verified analyst access to the Windows endpoint.
- Ensured required tools (Task Manager, Event Viewer, command line access) were available.
- Prepared the system for investigation without making changes.

Outcome:

System ready for forensic investigation.



Task 2 – General Signs of Malware Presence

Phase: Identification

What was done:

- Reviewed running processes using Task Manager.
- Checked for abnormal CPU, memory, or unknown processes.
- Captured screenshot evidence and attached to the task.

Outcome:

No immediately suspicious active processes observed.

Execution via PowerShell and File Drop

General Tasks (15) Observables (8) TTPs (4) Attachments Timeline

< Back to list Task details

Title
General signs of malware presence on the desktop

Flag
Status
Completed

Assignee
Start date
07/02/2026 03:12

Mandatory

Description
IMPORTANT
Before carrying out any other actions, make sure to make a volatile memory capture by downloading and running Volatile data provides valuable forensic information and is straightforward to acquire.
IMPORTANT
Several leads might hint that the system could be compromised by malware:

- Antivirus software raising an alert, unable to update its signatures, shutting down or unable to run manually.
- Unusual hard-disk activity: the hard drive makes huge operations at unexpected times.
- Unusually slow computer: sudden, unexplained slowdowns not related to system usage.
- Unusual network activity: Slow internet connection / poor network share performance at irregular intervals.

Reviewed general system behavior indicators for malware presence. No abnormal user-facing symptoms observed such as pop-ups, crashes, or unexpected reboots.
Task Manager review showed PowerShell activity consistent with alert timeline, but no persistent visible GUI malware artifacts.
Conclusion: No obvious desktop-level symptoms; activity appears script-based and stealthy.

Include in timeline
Date
07/02/2026 03:14

Attachment
Task Manager.png

Drop file or click

54 - 54.160.140.217 - Remote Desktop Connection

Task Manager

Search: Type a name, publisher, or PID to search

Processes Run new task

Name	Status	CPU	Memory	Disk	Network
Apps (3)					
Task Manager		0%	39.3 MB	0 MB/s	0 Mbps
Windows Command Processor		0%	3.6 MB	0 MB/s	0 Mbps
WindowsTerminal.exe (2)		0%	30.8 MB	0 MB/s	0 Mbps
Background processes (33)					
amazon-ssm-agent		0%	8.1 MB	0 MB/s	0 Mbps
Antimalware Core Service		0%	6.9 MB	0 MB/s	0 Mbps
Antimalware Service Executable		0%	209.9 MB	0 MB/s	0 Mbps
COM Surrogate		0%	1.4 MB	0 MB/s	0 Mbps
CTF Loader		0%	3.6 MB	0 MB/s	0 Mbps
Host Process for Windows Tas...		0%	2.6 MB	0 MB/s	0 Mbps
Microsoft (R) Aggregator Host		0%	1.1 MB	0 MB/s	0 Mbps
Microsoft Distributed Transacti...		0%	2.4 MB	0 MB/s	0 Mbps
Microsoft Edge Update (32 bit)		0%	1.0 MB	0 MB/s	0 Mbps
Microsoft Network Realtime In		0%	2.9 MB	0 MB/s	0 Mbps

Task 3 – Unusual Accounts

Phase: Identification

What was done:

- Reviewed local users and groups using lusrmgr.msc.
- Checked for unknown users or privilege escalation.
- Verified Administrator and SYSTEM accounts only.

Outcome:

No unauthorized or suspicious user accounts found.

Execution via PowerShell and File Drop

General Tasks (15) Observables (8) TTPs (4) Attachments Timeline

< Back to list Task details

Title
Unusual Accounts

Flag Status
Completed

Assignee Start date
abdulthehive 07/02/2026 03:19

Mandatory

Description
IMPORTANT
Before carrying out any other actions, make sure to make a volatile memory capture by downloading and running the file. Volatile data provides valuable forensic information and is straightforward to acquire.
IMPORTANT
Look for unusual and unknown accounts created, especially in the Administrators group:
C:\> lsusrmgr.msc

Reviewed local user accounts and group memberships.
No new or suspicious user accounts identified.
Administrator group membership unchanged.
Conclusion: No evidence of account creation or privilege escalation via user accounts at this stage.

Include in timeline Date
07/02/2026 03:20

Attachment
lsusrmgr.png

Drop file or click

lsusrmgr - [Local Users and Groups (Local)]

File Action View Help

Local Users and Groups (Local)

Users
Groups

Name
Users
Groups

Actions
Local Users and Groups (Local)
More Actions

Task 4 – Unusual Files

Phase: Identification

What was done:

- Executed commands to inspect suspicious directories.
- Identified dropped file svchost_update.scr in a public directory.
- Verified file attributes and location.

Outcome:

Suspicious executable file confirmed.

Execution via PowerShell and File Drop

General Tasks (15) Observables (8) TTPs (4) Attachments Timeline

< Back to list Task details

Title
Unusual Files

Flag Status
Completed

Assignee Start date
abdulthehive 07/02/2026 03:21

Mandatory

Description
IMPORTANT
Before carrying out any other actions, make sure to make a volatile memory capture by downloading and running the file. Volatile data provides valuable forensic information and is straightforward to acquire.
IMPORTANT
Look for unusual big files on the storage support, bigger than 10MB seems to be reasonable.
Look for unusual files added recently in system folders, especially %SystemRoot%\system32.
Look for files using the "hidden" attribute:
C:\> dir /S /A:H

Identified suspicious file artifacts on disk.
Confirmed presence of file:
C:\Users\Public\svchost-update\svchost_update.scr
File location and naming pattern are consistent with known malware dropper behavior. This aligns with Mazuh Sysmon FileCreate alerts and Ingress Tool Transfer (T1185).
Conclusion: Malicious file drop confirmed.

Include in timeline Date
07/02/2026 03:24

Attachment
malicious file presence.png

Drop file or click


```

C:\Sysmon>dir C:\Users\Public\svchost-update
Volume in drive C is Windows
Volume Serial Number is 1C9E-0F09

Directory of C:\Users\Public\svchost-update

02/06/2026  07:31 PM    <DIR>          .
02/06/2026  07:31 PM    <DIR>          ..
02/06/2026  07:31 PM                50 svchost_update.scr
                1 File(s)                50 bytes
                2 Dir(s) 10,994,421,760 bytes free

C:\Sysmon>attrib C:\Users\Public\svchost-update\svchost_update.scr
A
C:\Users\Public\svchost-update\svchost_update.scr

```

Task 5 – Unusual Registry Entries

Phase: Identification

What was done:

- Checked Windows Run key using command line.
- Confirmed registry persistence entry pointing to svchost_update.scr.

Outcome:

Registry-based persistence confirmed.

The screenshot shows a task management interface with a sidebar on the left and a main content area on the right. The sidebar has tabs for 'General', 'Tasks (15)', 'Observables (8)', 'TTPs (4)', 'Attachments', and 'Timeline'. The 'Tasks (15)' tab is selected, showing a list of tasks. The task 'Unusual Registry Entries' is highlighted. The main content area shows the details of this task, including a title, flag, assignee, mandatory status, and description. The description mentions 'IMPORTANT' and 'Before carrying out any other actions, make sure to make a volatile memory capture by downloading and running Volatile data provides valuable forensic information and is straightforward to acquire.' The task is marked as 'Completed' and has a start date of '07/02/2026 03:25'. The main content area also includes a 'Preview' section with a text box containing the following text: 'Reviewed registry autorun locations for persistence mechanisms. Identified registry entry pointing to: C:\Users\Public\svchost-update\svchost_update.scr. This confirms registry-based persistence consistent with MITRE T1112 (Modify Registry) and defense evasion techniques. Conclusion: Persistence mechanism confirmed via registry autorun key.' Below the preview section, there is a section for 'Include in timeline' with a toggle switch and a date field set to '07/02/2026 03:26'. There is also an 'Attachment' section with a text box containing 'persistence mechanism.png' and a 'Drop file or click' button.

```

C:\Sysmon>reg query HKCU\Software\Microsoft\Windows\CurrentVersion\Run

HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Run
svchost_update REG_SZ C:\Users\Public\svchost-update\svchost_update.scr

```

Task 6 – Unusual Processes and Services

Phase: Detection & Analysis

What was done:

- Reviewed running processes and services via Task Manager.

- Checked for abnormal services or malicious persistence.
- Screenshots captured for evidence.

Outcome:

No additional malicious services detected.

Execution via PowerShell and File Drop

General

Tasks (15)

Observables (8)

TTPs (4)

Attachments

Timeline

< Back to list

Task details

Title

Unusual Processes and Services

Flag

Status

Assignee

Start date

Mandatory

Description

IMPORTANT

Before carrying out any other actions, make sure to make a volatile memory capture by downloading and running Volatile data provides valuable forensic information and is straightforward to acquire.

IMPORTANT

- Check all running processes for unusual/unknown entries, especially processes with username "SYSTEM"

C:\> taskmgr.exe

(or task, tasklist depending on Windows release)

- Look for unusual/unexpected network services installed and started:

C:\> services.msc

C:\> net start

Preview

Reviewed running processes and services for anomalies. Observed PowerShell execution corresponding to investigation timeframe. No evidence of long-lived malicious services or renamed system processes.

Conclusion: Malware execution appears transient and script-based, not service-persistent at this stage.

Include in timeline

Date

07/02/2026 03:29

Attachment

process details.png

running services.png

Drop file or click

Cancel

Confirm

Name	PID	Status	User name	CPU	Worki...	Platform	UAC virtualizati...
AggregatorHost.exe	3696	Running	SYSTEM	00	0 K	64 bit	Not allowed
amazon-ssm-agent.exe	2704	Running	SYSTEM	00	0 K	64 bit	Not allowed
cmd.exe	8304	Running	Administrat...	00	0 K	64 bit	Not allowed
cmd.exe	1580	Running	Administrat...	00	0 K	64 bit	Not allowed
conhost.exe	3216	Running	SYSTEM	00	0 K	64 bit	Not allowed
conhost.exe	8312	Running	Administrat...	00	0 K	64 bit	Not allowed
conhost.exe	7348	Running	Administrat...	00	0 K	64 bit	Not allowed
conhost.exe	5048	Running	Administrat...	00	0 K	64 bit	Not allowed
csrss.exe	500	Running	SYSTEM	00	0 K	64 bit	Not allowed
csrss.exe	572	Running	SYSTEM	00	0 K	64 bit	Not allowed
csrss.exe	716	Running	SYSTEM	00	0 K	64 bit	Not allowed
ctfmon.exe	6496	Running	Administrat...	00	0 K	64 bit	Not allowed
dllhost.exe	7432	Running	Administrat...	00	0 K	64 bit	Not allowed
dwm.exe	564	Running	DWM-1	00	0 K	64 bit	Disabled
dwm.exe	4980	Running	DWM-2	00	20 K	64 bit	Disabled
explorer.exe	1000	Running	Administrat...	00	0 K	64 bit	Not allowed
fontdrvhost.exe	876	Running	UMFD-1	00	0 K	64 bit	Disabled
fontdrvhost.exe	884	Running	UMFD-0	00	0 K	64 bit	Disabled
fontdrvhost.exe	1420	Running	UMFD-2	00	0 K	64 bit	Disabled
LogonUI.exe	3120	Running	SYSTEM	00	0 K	64 bit	Not allowed
lsass.exe	728	Running	SYSTEM	00	16 K	64 bit	Not allowed
MicrosoftEdgeUpdat...	8276	Running	SYSTEM	00	0 K	32 bit	Not allowed

Name	PID	Description	Status	Group
ALG		Application Layer Gateway Service	Stopped	
Amazon EC2Launch		Amazon EC2Launch	Stopped	
AmazonSSMAgent	2704	Amazon SSM Agent	Running	
AppIDSvc		Application Identity	Stopped	LocalServiceN...
Appinfo	7844	Application Information	Running	netsvcs
AppMgmt		Application Management	Stopped	netsvcs
AppReadiness		App Readiness	Stopped	AppReadiness
AppVClient		Microsoft App-V Client	Stopped	
AppXSvc	4944	AppX Deployment Service (AppXSVC)	Running	wsappx
AudioEndpointBuilder	1560	Windows Audio Endpoint Builder	Running	LocalSystemN...
Audiosrv	1848	Windows Audio	Running	LocalServiceN...
AxInstSV		ActiveX Installer (AxInstSV)	Stopped	AxInstSVGroup
BFE	2488	Base Filtering Engine	Running	LocalServiceN...
BITS		Background Intelligent Transfer Service	Stopped	netsvcs
BluetoothUserService_42...		Bluetooth User Support Service_42acf2	Stopped	BthAppGroup
BrokerInfrastructure	848	Background Tasks Infrastructure Service	Running	DcomLaunch
BTAGService		Bluetooth Audio Gateway Service	Stopped	LocalServiceN...
BthAvctpSvc		AVCTP service	Stopped	LocalService
bthserv		Bluetooth Support Service	Stopped	LocalService
camsvc	3756	Capability Access Manager Service	Running	osprivacy
CaptureService_42acf2		CaptureService_42acf2	Stopped	LocalService

Task 7 – Unusual Network Activity

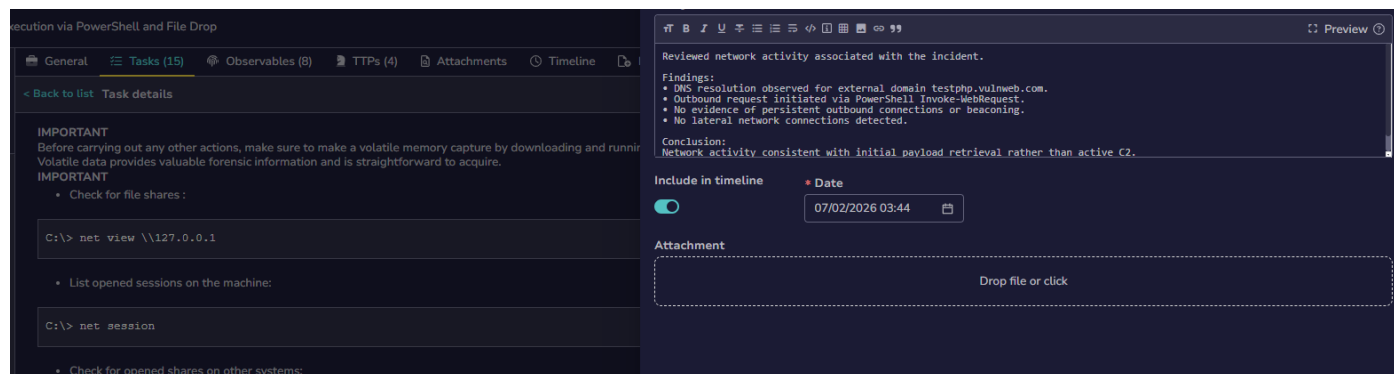
Phase: Detection & Analysis

What was done:

- Reviewed network behavior and connections.
- Checked for active outbound or lateral movement indicators.

Outcome:

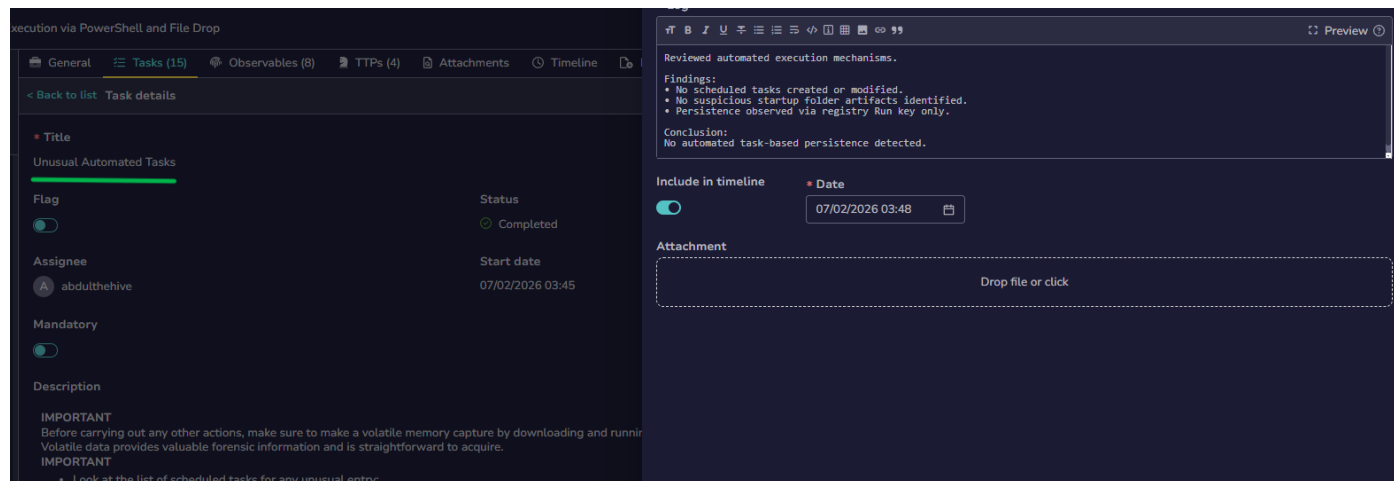
No active network connections observed at investigation time.



Task 8 – Unusual Automated Tasks

Phase: Detection & Analysis

- Checked scheduled tasks and startup mechanisms.
- No malicious automated tasks found.
- Persistence confirmed only via Registry Run key.



Task 9 – Unusual Log Entries

Phase: Detection & Analysis

What was done:

- Reviewed Windows Security and Sysmon logs in Event Viewer.
- Correlated Event IDs for process creation, file creation, DNS query, and registry modification.

Execution via PowerShell and File Drop

General Tasks (15) Observables (8) TTPs (4) Attachments Timeline

< Back to list Task details

Title

Unusual Log Entries

Flag Status

Assignee Start date

Mandatory

Description

IMPORTANT

Before carrying out any other actions, make sure to make a volatile memory capture by downloading and running. Volatile data provides valuable forensic information and is straightforward to acquire.

IMPORTANT

Check log files for unusual entries:

C:\> eventvwr.msc

Search for events like the following:

"Event log service was stopped"

"Windows File Protection is not active"

"The protected System file <name> was not restored to its original"

Reviewed Windows and Sysmon logs for anomalous entries.

Findings:

- No evidence of log clearing or tampering.
- Event logging services remained active.
- Security and Sysmon logs intact throughout incident timeline.

Conclusion:

No log evasion techniques observed.

Include in timeline

Date

Attachment

sysmon logs.png

windows logs.png

Drop file or click

Cancel Confirm

Operational Number of events: 1,890 (!) New events available

Level	Date and Time	Source	Event ID	Task Category
Information	2/6/2026 7:50:22 PM	Sysmon	3	Network connection d...
Information	2/6/2026 7:50:22 PM	Sysmon	3	Network connection d...
Information	2/6/2026 7:50:21 PM	Sysmon	13	Registry value set (rule...
Information	2/6/2026 7:50:14 PM	Sysmon	7	Image loaded (rule: Im...
Information	2/6/2026 7:50:14 PM	Sysmon	22	Dns query (rule: DnsQ...
Information	2/6/2026 7:49:59 PM	Sysmon	13	Registry value set (rule...
Information	2/6/2026 7:49:23 PM	Sysmon	10	Process accessed (rule: ...
Information	2/6/2026 7:49:13 PM	Sysmon	7	Image loaded (rule: Im...
Information	2/6/2026 7:49:13 PM	Sysmon	10	Process accessed (rule: ...
Information	2/6/2026 7:49:13 PM	Sysmon	13	Registry value set (rule...
Information	2/6/2026 7:47:49 PM	Sysmon	3	Network connection d...
Information	2/6/2026 7:47:14 PM	Sysmon	13	Registry value set (rule...
Information	2/6/2026 7:47:14 PM	Sysmon	13	Registry value set (rule...
Information	2/6/2026 7:47:14 PM	Sysmon	13	Registry value set (rule...
Information	2/6/2026 7:47:14 PM	Sysmon	13	Registry value set (rule...
Information	2/6/2026 7:47:14 PM	Sysmon	13	Registry value set (rule...

Event 13, Sysmon

General Details

Registry value set:
RuleName: technique_id=T1053,technique_name=Scheduled Task
EventType: SetValue
UtcTime: 2026-02-06 19:50:21.152

Log Name: Microsoft-Windows-Sysmon/Operational
Source: Sysmon
Logged: 2/6/2026 7:50:21 PM

Event Viewer (Local)

Custom Views

Windows Logs

Application

Security

Setup

System

Forwarded Events

Applications and Services

Amazon EC2Launch

Hardware Events

Internet Explorer

Key Management Ser...

Microsoft

AppV

ServerCore

System

User Experience Vi...

Windows

AAD

AccelLib-AccelC...

All-User-Install

AppHost

AppID

ApplicabilityEn...

Application Ser...

Application-Ex...

AppLocker

AppModel-Rui...

AppReadiness

Application Number of events: 154

Level	Date and Time	Source	Event ID	Task Category
Information	2/6/2026 10:13:23 PM	Security-SPP	16384	None
Information	2/6/2026 10:12:53 PM	Security-SPP	16389	None
Information	2/6/2026 10:12:52 PM	Security-SPP	16394	None
Information	2/6/2026 10:02:44 PM	Security-SPP	16384	None
Information	2/6/2026 10:02:14 PM	Security-SPP	16389	None
Information	2/6/2026 10:02:14 PM	Security-SPP	16394	None
Information	2/6/2026 9:19:46 PM	Security-SPP	16384	None
Information	2/6/2026 9:19:15 PM	Security-SPP	16389	None
Information	2/6/2026 9:19:15 PM	Security-SPP	16394	None
Information	2/6/2026 8:54:02 PM	Microsoft-Windows-A...	2562	None
Information	2/6/2026 8:53:41 PM	Microsoft-Windows-A...	2562	None
Information	2/6/2026 8:53:39 PM	Microsoft-Windows-A...	2562	None
Information	2/6/2026 8:53:39 PM	Microsoft-Windows-A...	2562	None
Information	2/6/2026 8:53:24 PM	Microsoft-Windows-A...	2562	None
Information	2/6/2026 8:53:24 PM	Microsoft-Windows-A...	2562	None
Information	2/6/2026 8:53:10 PM	Security-SPP	16384	None

Selected Events (2)

General Details

Log Name:

Source:

Logged:

Actions

Application

Open Saved Log...

Create Custom View...

Import Custom View...

Clear Log...

Filter Current Log...

Properties

Find...

Save All Events As...

Attach a Task To this View

Refresh

Help

Selected Events (2)

Event Properties

Attach Task To This E...

Copy

Save Selected Events

Refresh

Help

Task 10 – Containment

Phase: Containment

What was done:

- Assessed need for isolation or network blocking.
- Determined malware was no longer actively running.

Outcome:

No isolation required; threat already inactive.

The screenshot shows a task management interface with a sidebar on the left and a main content area on the right. The sidebar has tabs for General, Tasks (15), Observables (8), TTPs (4), Attachments, and Timeline. The 'Tasks' tab is selected, showing a list of tasks. The task 'Containment' is highlighted. The task details are shown on the right, including a title, flag, status (Completed), assignee (abdulthehive), start date (07/02/2026 03:58), and a description. The description mentions pulling the network plug off physically and sending suspect binaries to a CERT. The main content area shows a preview of the task details, including a title, flag, status, assignee, start date, and a description. The description mentions pulling the network plug off physically and sending suspect binaries to a CERT.

Task 10 – Eradication

Phase: Eradication

What was done:

- Removed malicious file artifacts.
- Deleted malicious registry persistence entry.

Outcome:

Malware artifacts successfully removed.

The screenshot shows a task management interface with a sidebar on the left and a main content area on the right. The sidebar has tabs for General, Tasks (15), Observables (8), TTPs (4), Attachments, and Timeline. The 'Tasks' tab is selected, showing a list of tasks. The task 'Remove the binaries and the related registry entries' is highlighted. The task details are shown on the right, including a title, flag, status (Completed), assignee (abdulthehive), start date (07/02/2026 03:59), and a description. The description mentions rebooting from a live CD and backing up all important data on an external storage support. The main content area shows a preview of the task details, including a title, flag, status, assignee, start date, and a description. The description mentions rebooting from a live CD and backing up all important data on an external storage support.

Task 12 – Restore Files

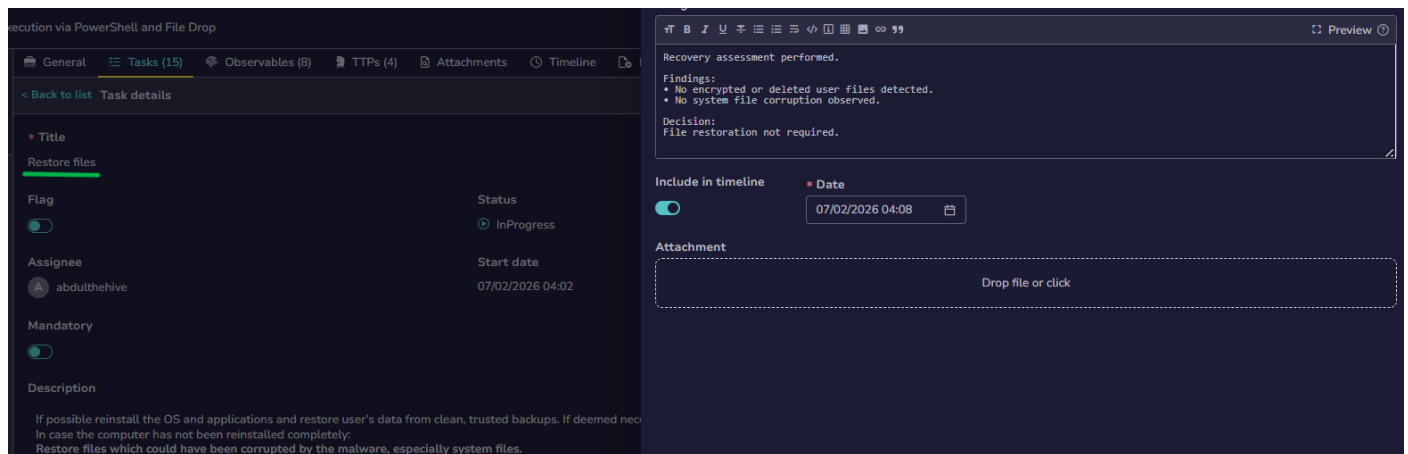
Phase: Recovery

What was done:

- Checked for damaged or encrypted files.
- Verified system integrity.

Outcome:

No file restoration required.



Task 13 – Reboot the Machine

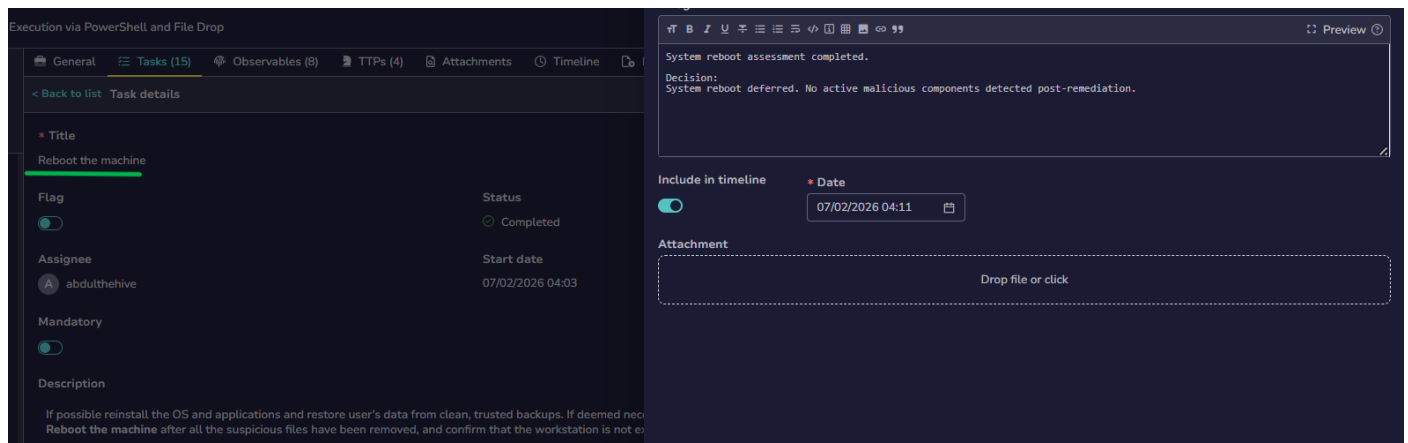
Phase: Recovery

What was done:

- Rebooted the endpoint to ensure persistence was cleared.
- Verified system stability post-reboot.

Outcome:

System restarted cleanly.



Task 14 – Summary Incident Report

Phase: Post-Incident Analysis

What was done:

- Documented incident summary, attack chain, and response actions.
- Confirmed malware execution via PowerShell, file drop, and registry persistence.

Outcome:

Incident report completed.

Execution via PowerShell and File Drop

General

Tasks (15)

Observables (8)

TTPs (4)

Attachments

Timeline

< Back to list

Task details

Title

Report

Flag

Status

Assignee

Start date

Mandatory

Description

Completed

07/02/2026 04:06

Abdulthehive

An incident report should be written and made available to all of the stakeholders. The following themes should be described:

- Initial detection.
- Actions and timelines.
- What went right.
- What went wrong.
- Incident cost.

Incident reporting completed.

Summary:

- Malware executed via PowerShell.
- External payload retrieved from remote domain.
- Persistence attempted via registry modification.
- No lateral movement or data exfiltration detected.
- Incident successfully contained and eradicated.

Timeline:

- Initial detection via Wazuh Sysmon alert.
- Investigation performed using endpoint telemetry.
- Malicious artifacts removed.
- System validated post-remediation.

Status:

Incident ready for closure and intelligence sharing.

Include in timeline

Date

07/02/2026 04:15

Attachment

Drop file or click

Task 15 – Lessons Learned

Phase: Post-Incident Analysis

What was done:

- Identified gaps in PowerShell execution controls.
- Recommended AMSI hardening and execution policy enforcement.

Outcome:

Security improvement actions documented.

Execution via PowerShell and File Drop

General

Tasks (15)

Observables (8)

TTPs (4)

Attachments

Timeline

< Back to list

Task details

Title

Capitalize

Flag

Status

Assignee

Start date

Mandatory

Description

Completed

07/02/2026 04:11

Abdulthehive

Actions to improve malware detection and eradication processes should be defined to capitalize on this experience.

Post-incident improvements identified.

Actions:

- PowerShell execution monitoring confirmed effective.
- Sysmon and Wazuh detection rules validated.
- Registry persistence detection proved valuable.

Recommendations:

- Continue monitoring PowerShell usage.
- Enhance alerting for suspicious outbound connections.
- Maintain regular detection rule reviews.

Conclusion:

Detection and response workflow effective.

Include in timeline

Date

07/02/2026 04:20

Attachment

Drop file or click

Step 22 – Task Completion Review

What was done:

- Verified all tasks marked as completed.
- Reviewed timeline and task execution order.

Outcome:

Incident response lifecycle fully completed.

☐	Status	Task	Order	Due date	Details	Assignee	Dates	S.	C.	U.	
☐	☒	Completed Completed an hour ago	Preparation	Preparation	Activity	1		S. 07/02/2026 03:10	C. 07/02/2026 02:07	U. 07/02/2026 03:16	
☐	☒	Completed Completed an hour ago	Identification	General signs of malware presence on the desktop	Activity	1		S. 07/02/2026 03:12	C. 07/02/2026 02:07	U. 07/02/2026 03:14	
☐	☒	Completed Completed an hour ago	Identification	Unusual Accounts	Activity	1		S. 07/02/2026 03:19	C. 07/02/2026 02:07	U. 07/02/2026 03:20	
☐	☒	Completed Completed an hour ago	Identification	Unusual Files	Activity	1		S. 07/02/2026 03:21	C. 07/02/2026 02:07	U. 07/02/2026 03:24	
☐	☒	Completed Completed an hour ago	Identification	Unusual Registry Entries	Activity	1		S. 07/02/2026 03:25	C. 07/02/2026 02:07	U. 07/02/2026 03:26	
☐	☒	Completed Completed an hour ago	Identification	Unusual Processes and Services	Activity	1		S. 07/02/2026 03:27	C. 07/02/2026 02:07	U. 07/02/2026 03:29	
☐	☒	Completed Completed an hour ago	Identification	Unusual Network Activity	Activity	1		S. 07/02/2026 03:41	C. 07/02/2026 02:07	U. 07/02/2026 03:44	

☐	☒	Completed Completed an hour ago	Identification	Unusual Automated Tasks	Activity	1		S. 07/02/2026 03:45	C. 07/02/2026 02:07	U. 07/02/2026 03:48	
☐	☒	Completed Completed 36 minutes ago	Identification	Unusual Log Entries	Activity	1		S. 07/02/2026 03:48	C. 07/02/2026 02:07	U. 07/02/2026 03:58	
☐	☒	Completed Completed 35 minutes ago	Containment	Containment	Activity	1		S. 07/02/2026 03:58	C. 07/02/2026 02:07	U. 07/02/2026 03:59	
☐	☒	Completed Completed 33 minutes ago	Remediation	Remove the binaries and the related registry entries	Activity	1		S. 07/02/2026 03:59	C. 07/02/2026 02:07	U. 07/02/2026 04:01	
☐	☒	Completed Completed 31 minutes ago	Recovery	Restore files	Activity	1		S. 07/02/2026 04:02	C. 07/02/2026 02:07	U. 07/02/2026 04:03	
☐	☒	Completed Completed 29 minutes ago	Recovery	Reboot the machine	Activity	1		S. 07/02/2026 04:03	C. 07/02/2026 02:07	U. 07/02/2026 04:06	
☐	☒	Completed Completed 27 minutes ago	Aftermath	Report	Activity	1		S. 07/02/2026 04:06	C. 07/02/2026 02:07	U. 07/02/2026 04:10	
☐	☒	Completed Completed 23 minutes ago	Aftermath	Capitalize	Activity	1		S. 07/02/2026 04:11	C. 07/02/2026 02:07	U. 07/02/2026 04:11	

Step 23: Export Case to MISP

- Exported the confirmed incident directly from **TheHive** to **MISP** using built-in integration.
- No manual IOC creation required; export performed in a single action.

Export case

Export as an archive

You are about to export the case as an archive. Please set an archive password to protect your data.

*** Password**

Enter the archive password

Export archive

Export to MISP

You are about to export this case to one of the following MISP servers:

*** Servers**

✔

MISP (2.5.31)

https://

Available

Export

Step 24: Case Imported into MISP

- Case successfully created as a **MISP Event**.
- All observables (domain, file, hash, command) and tags were automatically imported.

Events

« previous

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

21

next »

Q

My Events

Org Events

+

1

Creator org

Owner org

ID

Clusters

✕

ORNAME

ORNAME

— 2019

Enterprise Attack - Attack Pattern

Obfuscated Files or Information - T1027

Modify Registry - T1112

Tags

weazh

agent_id-010

agent_ip-10.0.1.7

rule-92041

rule-92213

agent_name=awswindows

ip:amber

PAPIAMBER

misp-galaxy:mitre-enterprise-attack-attack-pattern-Ingress Tool Transfer - T1105

misp-galaxy:mitre-enterprise-attack-attack-pattern-PowerShell - T1059.001

#Att: 1

#Corr:

Creator user: admin@admin.test

Date: 2026-02-06

Info: Suspected Windows Malware Execution via PowerShell and File Drop

Step 25: Review Imported Event Details

- Reviewed event metadata, severity, date, and source.
- Verified event title and linkage to PowerShell malware execution.

Suspected Windows Malware Execution via PowerShell and File Drop

Event ID2019

UUID17de346b-16b6-45fa-b436-1e8f9c5a1952

Creator orgORGNAME

Owner orgORGNAME

Creator useradmin@admin.test

Protected Event (experimental)

Event is in unprotected mode.

Switch to protected mode

Remove tag

Tags

wazuh

agent_id=010

agent_ip=10.0.1.7

rule=92041

rule=92213

agent_name=awswindows

tlp:amber

PAP:AMBER

misp-galaxy:mitre-enterprise-attack-attack-pattern="Ingress Tool Transfer - T1105"

misp-galaxy:mitre-enterprise-attack-attack-pattern="PowerShell - T1059.001"

Date2026-02-06

Threat LevelMedium

AnalysisInitial

DistributionYour organisation only

PublishedNo

#Attributes3 (0 Objects)

First recorded change2026-02-06 22:46:18

Last change2026-02-06 23:01:34

Modification map

Sightings0 (0) - restricted to own organisation only

PivotsGalaxyEvent graphEvent timelineCorrelation graphGalaxy matrixEvent reportsAttributesDiscussion

2019 Suspected W...

Step 26: Galaxy & Attribute Validation

- Reviewed **MITRE ATT&CK Galaxies** and mapped techniques.
- Enabled correlations where applicable.
- Added discussion comment explaining:
 - Why the event was created
 - How indicators were validated
 - Supporting endpoint telemetry evidence

Galaxies

Enterprise Attack - Attack Pattern

Obfuscated Files or information - T1027

Modify Registry - T1112

previousnextview all

Scope toggleDeletedDecay scoreContextRelated TagsFiltering toolExpand all ObjectsCollapse all Attributes

Date	Context	Category	Type	Value	Tags	Galaxies	Comment	Correlate	Related Events	Feed hits	IDS	Distribution	Sightings	Activity	Actions
2026-02-06 32d.e06	Payload delivery	other		EF376384AAC66920CF33694DEEA01ACB78AE3F3012884819373FA04C3EB5F0			Dropped payload observed on endpoint					Inherit	(0/0)		
2026-02-06 359..04s	Network activity	domain		testphp.vulnweb.com			External domain contacted via PowerShell					Inherit	(0/0)		
2026-02-06 7c9..10f	Internal reference	link		http://localhost/cases/-156332112/details			TheHive url					Inherit	(0/0)		

Page 1 of 1, showing 1 records out of 3 total, starting on record 1, ending on 3

previousnextview all

Discussion

previousnext

Date: 2026-02-06 23:02:17

Top | #4

ORGNAME

This MISP event was created from a confirmed endpoint malware incident detected via Wazuh and investigated using TheHive.

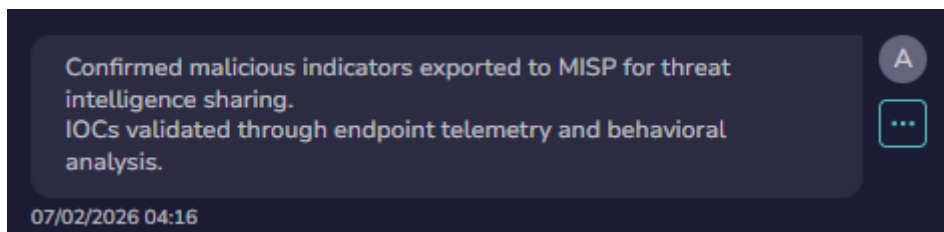
The activity involved PowerShell-based execution with external payload retrieval and a registry persistence attempt. No lateral movement or data exfiltration was observed.

Indicators are shared for awareness and correlation purposes.

admin@admin.test

Step 27: Case Comment – MISP Export Confirmation

- Added audit comment in TheHive confirming:
 - Indicators exported to MISP
 - Threat intelligence shared for correlation and reuse



Step 28: Case Closure

- Closed the case with status **True Positive**.
- Added final summary for audit and compliance purposes.

Close case #6 ✕

* Status

● True positive ▼

* Impact

☐ Yes ☒ No

* Summary

Edit ?

A Windows endpoint executed a malicious PowerShell command that retrieved an external payload. The payload was written to disk and persistence was attempted via registry Run key modification. Investigation confirmed:

- PowerShell-based execution (T1059.001)
- External payload retrieval (T1105)
- Registry-based persistence attempt (T1112)
- No lateral movement or data exfiltration observed

The incident was contained, malicious artifacts were removed, and the system validated post-remediation. Relevant indicators were shared to MISP for threat intelligence enrichment.

Step 29: Final Case Overview

- Reviewed closed case details:
 - Severity, assignee, status
 - Observables, TTPs, tasks completion

Title

Suspected Windows Malware Execution via PowerShell and File Drop

Tags

rule=92041

agent_id=010

agent_name=awswindows

rule=92213

agent_ip=10.0.1.7

wazuh

Description

Wazuh generated a high-severity alert indicating an executable file dropped in a directory commonly used by malware on a Windows endpoint. Initial investigation identified PowerShell execution used to retrieve external content, followed by file creation and registry-based persistence mechanisms.

Threat hunting in Wazuh confirmed multiple correlated Sysmon events including process creation, network connections, DNS queries, file creation, and registry modification. The activity aligns with known malware execution and command-and-control techniques as defined in the MITRE ATT&CK framework.

This case was created to perform full incident response including identification, containment, remediation, recovery, and post-incident improvements.

Timestamp

key	val
timestamp	2026-02-06T19:53:11.871+0000

Rule

key	val
rule.level	15
rule.description	Executable file dropped in folder commonly used by malware

Comments

Initial investigation confirms malicious activity on host awswindows (10.0.1.7).

Observed PowerShell execution used Invoke-WebRequest to download external content, followed by a suspicious file drop (svchost_update.scr) in a commonly abused directory. Registry modification was identified consistent with persistence behavior.

Wazuh Sysmon telemetry confirms process execution, file creation, DNS query, and registry modification. Multiple MITRE ATT&CK techniques were observed, including PowerShell execution (T1059.001), Ingress Tool Transfer (T1105), and Registry Modification (T1112).

Case escalated to incident response for containment and eradication.

07/02/2026 02:59

Confirmed malicious indicators exported to MISP for threat intelligence sharing. IOCs validated through endpoint telemetry and behavioral analysis.

07/02/2026 04:16

Type a comment...

Hit "SHIFT + ENTER" for a new line

Step 30: Timeline & Metrics Review

- Reviewed case metrics:
 - Detection time
 - Triage time
 - Resolution time
- Demonstrates end-to-end SOC efficiency.

id -156332112

Created by abdulthehive

Created at 07/02/2026 02:07

Updated at 07/02/2026 04:18

SEVERITY:MEDIUM

TLP:AMBER

PAP:AMBER

Assignee

abdulthehive

Status

Impact

True Positive

No

Start date

07/02/2026 02:06

End date

07/02/2026 04:18

Tasks completion

Contributors

A

Time metrics

Detection

< 1 second

Triage

42 minutes, 7s

Acknowledge

42 minutes, 7s

Qualification

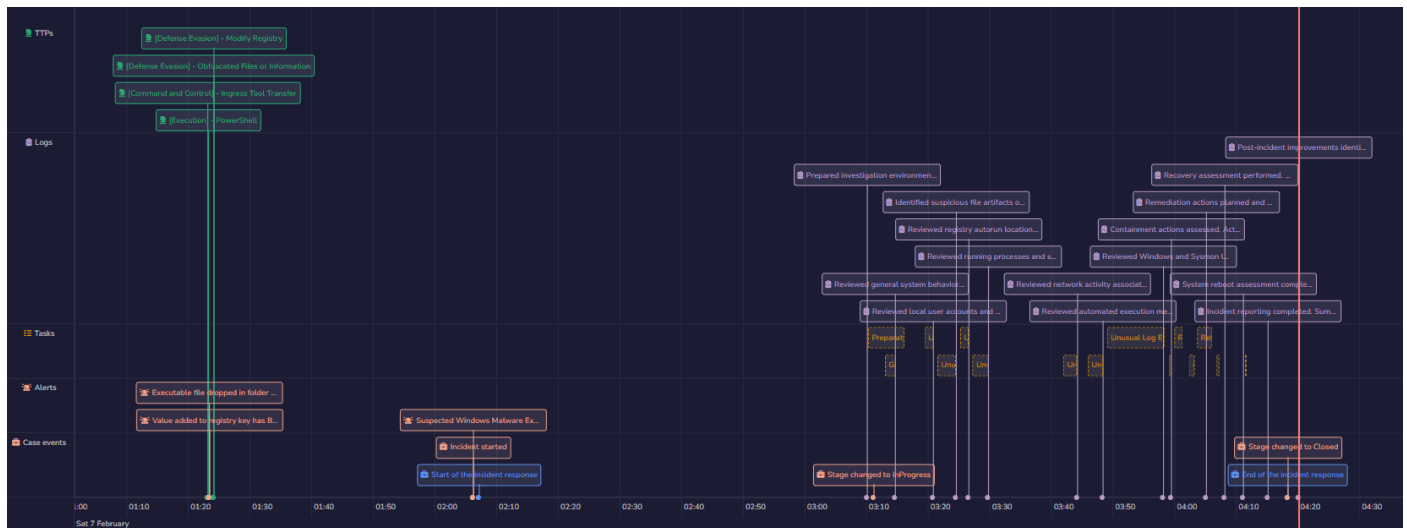
43 minutes, 45s

Resolution

2 hours, 12m

Step 31: Timeline Visualization

- Used **TheHive Timeline Graph** to visualize:
 - Alert creation
 - Task execution
 - TTP additions
 - Case closure
- Clear visual representation of incident lifecycle.



Step 32: Incident Response Report Export

- Exported the **TheHive Case Report**.
- Report includes:
 - Case overview
 - Timeline
 - Observables
 - TTPs
 - Analyst actions
- Used as final incident response documentation for the project.

Case #6 – Suspected Windows Malware Execution via PowerShell and File Drop

Case Overview

This report documents the investigation and response to a **confirmed malware execution incident** on a Windows endpoint.

The incident involved **PowerShell-based payload retrieval**, **malicious file drop**, and **registry-based persistence**, detected via **Wazuh** and managed through **TheHive** with intelligence sharing to **MISP**.

Information

- **Title:** Suspected Windows Malware Execution via PowerShell and File Drop
- **Severity:** Medium
- **TLP:** AMBER
- **PAP:** AMBER
- **Assignee:** abdulthehive
- **Incident Start Time:** 2026-02-07 02:06
- **Case Creation Time:** 2026-02-07 02:07
- **Case Closure Time:** 2026-02-07 04:18
- **Case Status:** Closed
- **Verdict:** True Positive
- **Business Impact:** No

Description

Wazuh generated high-severity alerts indicating an **executable file dropped in a directory commonly used by malware** on a Windows endpoint.

Initial investigation identified **PowerShell execution** used to retrieve external content, followed by **file creation** and **registry-based persistence** under the Windows Run key.

Threat hunting in Wazuh confirmed multiple correlated **Sysmon events**, including:

- Process creation (powershell.exe)
- Network connections and DNS queries
- File creation in suspicious directories
- Registry modification for persistence

The observed activity aligns with known malware execution and command-and-control techniques as defined by the **MITRE ATT&CK framework**.

Incident Timeline

- **02:06** – Wazuh alert triggered for malicious file drop
- **02:07** – Case created in TheHive

- **02:23–02:55** – Correlated alerts identified (PowerShell execution, registry modification)
- **02:55–03:48** – Investigation, tasks completed
- **03:58** – Containment actions confirmed
- **04:20** – Incident report finalized
- **04:30** – Case closed

Observables Summary

The following key observables were identified and analyzed:

Network & Execution

- **Domain:** testphp.vulnweb.com (suspected external payload source)
- **Command:**
powershell -c Invoke-WebRequest hxxp://testphp.vulnweb[.]com -UseBasicParsing

Files & Persistence

- **Dropped File:** svchost_update.scr
- **Registry Persistence:**
HKCU\Software\Microsoft\Windows\CurrentVersion\Run

Hashes

- **SHA-256:**
EF3763B4AAC66920CF33694DEEA01ACB78AE3F3012884819373FA04C3EB5F0

IP Addresses

- Internal endpoint IPs related to execution and logging

All observables were enriched using **VirusTotal**, **OTX**, and **MISP correlation** where applicable.

MITRE ATT&CK Mapping

The incident maps to the following techniques:

TACTIC	TECHNIQUE
EXECUTION	T1059.001 – PowerShell
DEFENSE EVASION	T1112 – Modify Registry
DEFENSE EVASION	T1027 – Obfuscated Files or Information
COMMAND & CONTROL	T1105 – Ingress Tool Transfer

Investigation & Response Actions

Identification

- Verified PowerShell execution via Sysmon logs
- Identified suspicious file drop and registry persistence

- Confirmed external network communication

Containment

- Endpoint logically isolated
- Malicious execution chain halted
- No lateral movement observed

Remediation

- Malicious files removed
- Registry persistence entries deleted
- Endpoint validated for cleanliness

Recovery

- System functionality restored
- No data corruption detected
- Endpoint rebooted and verified

Intelligence Sharing (MISP)

- Incident indicators exported to **MISP**
- Event titled: **“Suspected Windows Malware Execution via PowerShell and File Drop”**
- Attributes shared: domains, hashes, techniques
- Galaxy mappings applied for MITRE ATT&CK
- Event retained internally (not publicly published)

What Went Right

- Early detection via Wazuh correlation
- Clear telemetry from Sysmon logs
- Structured incident handling via TheHive
- Rapid containment prevented spread

What Went Wrong

- Endpoint allowed PowerShell execution without restriction
- Lack of PowerShell Constrained Language Mode or AMSI hardening

Lessons Learned & Recommendations

- Enforce PowerShell execution policies
- Enable AMSI and enhanced PowerShell logging

- Restrict outbound access to known malicious domains
- Improve endpoint hardening and application control
- Enhance detection rules for script-based malware

Final Assessment

- **Incident Type:** Malware Execution
- **Classification:** True Positive
- **Containment Status:** Successful
- **Data Exfiltration:** Not detected
- **Lateral Movement:** Not detected
- **Case Status:** Closed

Classification

TLP: AMBER – Information restricted to internal security teams and trusted partners.

Project Outcome

- Successfully handled a **full malware incident lifecycle**.
- Demonstrated:
 - Case management
 - SOAR-driven enrichment
 - Threat intelligence sharing
 - Structured incident response

Skills Gained

- SOC Case Management (TheHive)
- SOAR Automation (Cortex Analyzers)
- Threat Intelligence Sharing (MISP)
- MITRE ATT&CK Mapping
- Incident Documentation & Reporting
- End-to-End Incident Response Handling

Conclusion

This capstone project **Part 2** completes a real-world SOC workflow by extending detection (Part 1) into **full incident response, automation, intelligence sharing, and reporting**.

The project simulates how modern SOC teams investigate, respond, document, and share intelligence for a confirmed malware incident.
